

Восстановительный ЖГГ:

$$\begin{aligned} N_{\text{ВГГ}}(\pi_{\text{T}}) &= \dot{m}_{\text{ВГГ}} * R_{\text{ВГГ}} * T_{\text{ГГ}} * \eta_{\text{T}} * \left(1 - \frac{1}{\pi_{\text{T}}^{\frac{(k_{\text{ГГ}}-1)}{k_{\text{ГГ}}}}} \right) * \frac{k_{\text{ГГ}}}{(k_{\text{ГГ}} - 1)} = \\ &= 3.837 * 2500 * 700 * 0,6 * \left(1 - \frac{1}{\pi_{\text{T}}^{0,23}} \right) * 4,33 = \\ &= 17,445 * 10^6 * \left(1 - \frac{1}{\pi_{\text{T}}^{0,23}} \right) \text{ Вт.} \end{aligned}$$

Мощность насосов:

$$\begin{aligned} N_{\text{H}}(\pi_{\text{T}}) &= Q_{\Sigma} * \frac{[\pi_{\text{T}} * (p_{\text{K}} + \Delta p_{\text{K}}) + (\Delta p_{\text{ЖГГ}} - p_{\text{H ВХ}})]}{\eta_{\text{H}}} = \\ &= 0,049 * \frac{[\pi_{\text{T}} * (5 + 1,5) + (3 - 0,5)] * 10^6}{0,65} = 0,075 * 10^6 * (6,5 * \pi_{\text{T}} + 2,5) \text{ Вт} \end{aligned}$$

$$N_{\text{H}}(\pi_{\text{T}}) = 0,075 * 10^6 * (6,5 * \pi_{\text{T}} + 2,5) \text{ Вт}$$

$$N_{\text{ОГГ}}(\pi_{\text{T}}) = 10,778 * 10^6 * \left(1 - \frac{1}{\pi_{\text{T}}^{0,23}} \right) \text{ Вт}$$

$$N_{\text{ВГГ}}(\pi_{\text{T}}) = 17,445 * 10^6 * \left(1 - \frac{1}{\pi_{\text{T}}^{0,23}} \right) \text{ Вт}$$